

FALCON: Supplementary Dataset, Implementation, and Reproducibility Details

Dataset Details

We use the *eICU Collaborative Research Database (eICU-CRD)* for in-hospital mortality prediction. We analyze **10 hospitals** to characterize cross-hospital heterogeneity, with **five hospitals (167, 199, 252, 420, and 458)** used as federated clients in the primary strict LOHO experiments.

As shown in Table 1, the hospitals exhibit substantial non-IID variation. Patient populations range from **2,097 patients at Hospital 283** to **6,092 patients at Hospital 167**, while mortality prevalence varies from **4.03% to 9.36%** across institutions. Demographic characteristics, including sex, ethnicity, and obesity prevalence, also differ considerably. In addition, variations in mean Lactate and Blood Urea Nitrogen (BUN) values indicate physiological covariate shifts across hospitals. These differences provide a realistic setting for evaluating FALCON under heterogeneous clinical distributions.

Input features include **demographics, laboratory measurements, medications, diagnosis and treatment information, and clinical time-series representations**. This table summarizes the demographic and clinical heterogeneity across the 10 hospitals.

Hospital	Patients	Mortality(%)	Male (%)	Caucasian (%)	Obese(%)	Mean Lactate	Mean BUN
167	6092	4.22	55.50	80.30	28.90	1.69	20.77
420	4679	9.36	58.10	93.60	34.40	1.73	24.27
199	4240	6.32	56.00	97.20	44.00	1.76	23.36
458	3701	9.05	55.10	69.50	37.30	1.83	22.60
252	3371	9.26	57.50	93.20	39.90	1.80	22.55
165	3053	4.42	53.60	87.30	30.30	1.69	20.04
148	2881	4.03	53.70	94.90	28.60	1.67	23.16
281	2531	5.29	53.10	84.60	39.50	1.67	23.65
449	2110	7.58	54.90	92.20	37.10	1.93	22.91
283	2097	6.01	50.90	95.20	32.00	1.62	26.45

Table 1. Demographic and clinical heterogeneity across hospitals

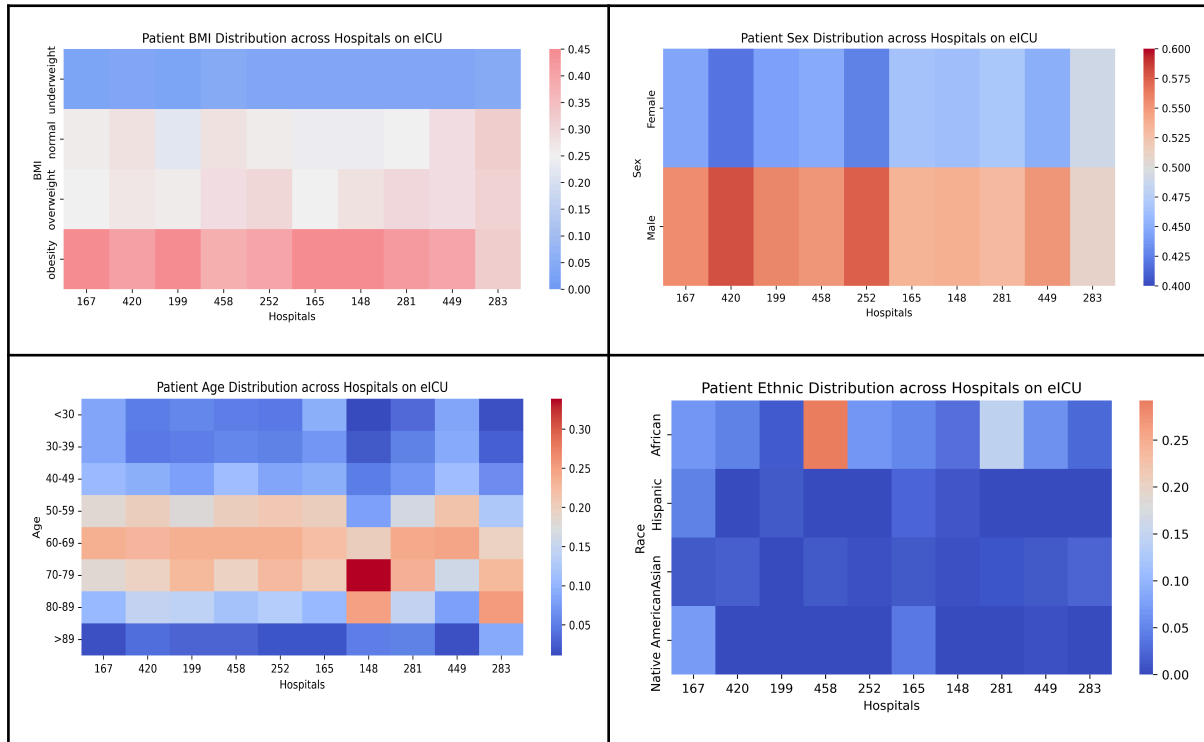
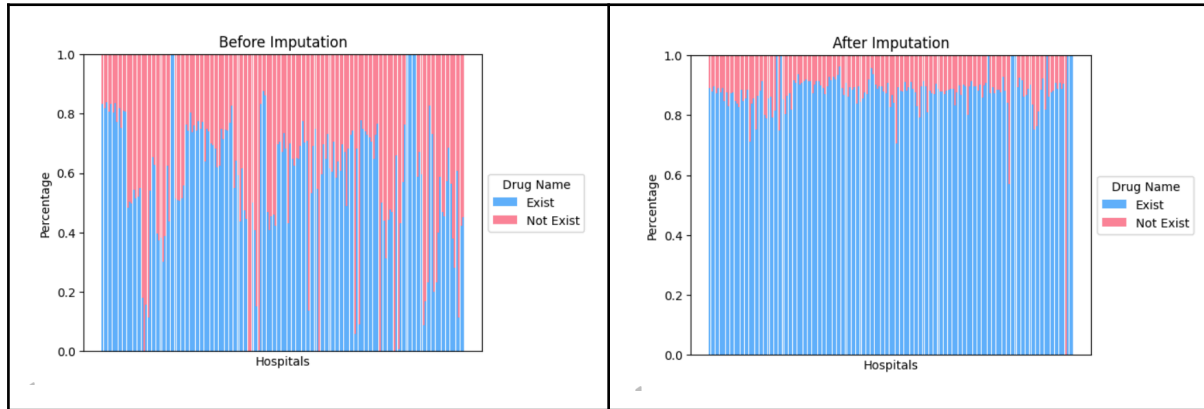


Figure 1. Demographic heterogeneity across the 10 analyzed hospitals based on age, BMI, and sex distributions.

Preprocessing Details

1. **Hospital and Patient Selection:** Records were restricted to the selected eICU hospitals and valid ICU stays. The prediction target was in-hospital mortality.
2. **48-Hour Observation Window:** Only clinical information available within the first 48 hours of ICU admission was used to construct patient features.
3. **Demographic Processing:** Demographic variables such as age, sex, race, and BMI were cleaned and converted into model-compatible categorical or numerical representations.
4. **Laboratory Feature Processing:** Selected laboratory measurements recorded within the first 48 hours were aggregated at the patient level. Missing laboratory values were imputed using the corresponding median values.
5. **Medication Processing:** Medication records were harmonized using available drug names and identifiers. Only medications administered or recorded within the first 48 hours of ICU admission were retained to prevent the inclusion of information from later stages of hospitalization.
6. **Clinical Indicator Extraction:** Sepsis and ventilation-related indicators were derived from the available diagnosis and treatment records.
7. **Diagnosis and Treatment Representations:** Diagnosis and treatment information was processed into structured feature representations. In the extended feature setting, their temporal patterns were further captured using time-series embeddings.

Final Feature Construction: The processed demographic, laboratory, medication, clinical indicator, and diagnosis/treatment features were combined into patient-level representations for federated training and evaluation.



Treatment Embeddings (for Ablation):

Treatment records were first organized hierarchically based on the available treatment categories. Each patient's treatment events within the observation window were mapped to discrete temporal bins, producing a multi-hot representation of treatment activity over time. To obtain compact patient-level representations, the resulting treatment sequences were processed using a neural embedding model that projected the high-dimensional temporal treatment patterns into a lower-dimensional latent space. The learned treatment embedding captures both the presence of specific treatment categories and their temporal evolution during the early ICU stay. These embeddings were subsequently concatenated with the remaining clinical features.

Diagnosis Embeddings (for Ablation):

Diagnosis records were similarly transformed into structured temporal sequences. Diagnosis events were mapped to diagnosis categories and assigned to temporal bins within the observation window, resulting in patient-specific multi-hot sequences. A neural representation model was then used to encode these sparse, high-dimensional sequences into fixed-length latent embeddings. The resulting diagnosis embeddings capture co-occurrence and temporal patterns of clinical conditions during the early ICU stay and were incorporated into the final patient-level feature representation.

General Experimental Settings

The following settings were used as the common experimental configuration for the RQ1 and RQ2 comparisons.

Parameter	Setting
Dataset	eICU Collaborative Research Database (eICU-CRD)
Prediction Task	In-hospital mortality prediction
Client Hospitals	167, 199, 252, 420, 458
Evaluation Protocol	Strict Leave-One-Hospital-Out (LOHO)
Non-personalized Methods Split	80% Training / 20% Testing

Parameter	Setting
Personalized Methods Split	60% Training / 20% Validation / 20% Testing
Federated Rounds	25
Local Epochs	4
Batch Size	256
Optimizer	AdamW
Base Learning Rate	1.5×10^{-3}
Learning Rate Schedule	Cosine Decay
Local Weight Decay	1×10^{-4}
Evaluation Metrics	AUROC, AUPRC, F1-score, Precision, Recall
Random Seed	42

For non-personalized federated methods, each client's data was divided into 80% training and 20% testing. For personalized methods, the data was divided into 60% training, 20% validation, and 20% testing. All methods were evaluated under the strict LOHO protocol, where each participating hospital was held out in turn as an unseen test hospital, and results are reported as mean $\pm$ standard deviation across the held-out hospitals.

HyperParameter Settings:

Hyperparams for FALCON model

Hyperparameter	Value
Random seed	42
Federated rounds	25
Local epochs	5
Batch size	256
Base learning rate	1.5×10^{-3}
Learning rate schedule	Cosine decay
Minimum LR fraction	0.1
Optimizer	AdamW
Local weight decay	1×10^{-4}
Encoder hidden dimension	64
Token dimension	16

Hyperparameter	Value
Number of ResNet blocks	2
Dropout	0.2
Focal loss γ	2.0
Supervised contrastive weight	0.05
Contrastive temperature	0.2
Prototype alignment weight	0.05
FedProx coefficient	1×10^{-4}
Gradient clipping	1.0
SWD projections	128
Maximum SWD samples	1024
SWD similarity parameter β	2.0
Meta-aggregator learning rate	1×10^{-4}
Meta-aggregator weight decay	1×10^{-2}
Aggregator temperature τ	2.0
Weight smoothing factor	0.10
Maximum client weight	0.35
Meta gradient clipping	0.5
Bootstrap iterations	100

FALCON was trained for 25 federated communication rounds, with five local epochs per round and a batch size of 256. Local optimization used AdamW with a base learning rate of 1.5×10^{-3} and cosine learning-rate decay. The Tabular ResNet encoder used a hidden dimension of 64, token dimension of 16, two residual blocks, and dropout of 0.2. Focal loss ($\gamma=2.0$), supervised contrastive learning (weight 0.05, temperature 0.2), prototype alignment (weight 0.05), and FedProx regularization ($\mu=10^{-4}$) were used during local training. SWD was computed using 128 random projections, while the convolutional meta-aggregator used a temperature of 2.0, smoothing factor of 0.10, and a maximum client weight of 0.35.